MATH 128 - HW9 - Jie Wen

Problem 1: Problems from the textbook, chapter 3.2

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$$P_{1,2} = \frac{(0.5-0.4)P_2 - (0.5-0.7)P_1}{0.7-0.4} = \frac{0.1P_2 + 0.56}{0.3}.$$

Use $P_{0,1,2} = \dfrac{(x-x_0)P_{1,2} - (x-x_2)P_{0,1}}{x_2 - x_0}$:

$$\frac{27}{7} = \frac{0.5P_{1,2} - (-0.2)\cdot 3.5}{0.7} = \frac{0.5P_{1,2} + 0.7}{0.7}.$$

Hence $P_{1,2} = 4$.

$$\frac{0.1P_2 + 0.56}{0.3} = 4 \Rightarrow 0.1P_2 + 0.56 = 1.2 \Rightarrow P_2 = 6.4.$$

So $f(0.7) = P_2 = 6.4$.

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From the given linear interpolants,

$$f(0) = P_{0,1}(0) = 1, \quad f(1) = P_{0,1}(1) = 3, \quad f(2) = P_{0,2}(2) = 3.$$

Use the Lagrange basis on $\{1,2,3\}$ to extract $f(3)$ from $P_{1,2,3}(2.5) = 3$:

$$P_{1,2,3}(2.5) = 3L_1(2.5) + 3L_2(2.5) + f(3)L_3(2.5),$$

with $L_1(2.5) = -0.125, L_2(2.5) = 0.75, L_3(2.5) = 0.375$ (standard Lagrange basis). Setting this equal to 3 gives

$$3 = 3(0.625) + 0.375f(3) \Rightarrow f(3) = 3.$$

Compute $P_{0,1,2}(2.5)$ (quadratic through $\{0,1,2\}$) via Lagrange:

$$P_{0,1,2}(2.5) = 1\cdot 0.375 + 3\cdot(-1.25) + 3\cdot 1.875 = 2.25.$$

Use Neville's recursion for the cubic (with $x = 2.5$):

$$P_{0,1,2,3}(x) = \frac{(x-x_0)P_{1,2,3}(x) - (x-x_3)P_{0,1,2}(x)}{x_3 - x_0}.$$

Thus

$$P_{0,1,2,3}(2.5) = \frac{(2.5-0)\cdot 3 - (2.5-3)\cdot 2.25}{3-0} = \frac{7.5 + 1.125}{3} = \frac{23}{8} = 2.875.$$

Problem 2: Divided Differences Matlab

(a)

```
function [a, F] = newton_divdiff(x, y)
    x = x(:);                    % ensure column
    y = y(:);
    n = numel(x) - 1;

    % basic checks
    if numel(y) ~= numel(x)
        error('x and y must have the same length.');
    end
    if numel(unique(x)) ~= numel(x)
        error('All nodes x_i must be distinct.');
    end

    % Initialize table: first column is zeroth divided differences f[x_i] = y_i
    F = zeros(n+1, n+1);
```

```
    F(:,1) = y;

    % Build higher-order divided differences (Algorithm 3.2)
    for j = 2:n+1               % column = order j-1
        for i = j:n+1           % row index
            F(i,j) = (F(i,j-1) - F(i-1,j-1)) / (x(i) - x(i-j+1));
        end
    end

    % Newton coefficients are the diagonal entries
    a = diag(F);
end
```

(b)

```
function p = newton_eval(xnodes, a, xq)
    xnodes = xnodes(:);
    a = a(:);
    N = numel(a);

    % Start from highest-order coefficient and nest down
    p = a(end) .* ones(size(xq));
    for k = N-1:-1:1
        p = a(k) + (xq - xnodes(k)) .* p;
    end
end
```

Problem 3: Divided differences

The divided difference $f[x_0, x_1, \ldots, x_k]$ is the leading coefficient of the unique interpolating polynomial $P(x)$ of degree at most k that agrees with f at the distinct points $x_0, x_1, \ldots, x_k$, expressed in Newton's divided-difference form starting with x_0:

$$P(x) = f[x_0] + f[x_0, x_1](x - x_0) + \cdots + f[x_0, \ldots, x_k](x - x_0) \cdots (x - x_{k-1}).$$

The coefficient of x^k in this expansion is $f[x_0, x_1, \ldots, x_k]$, since the leading coefficient of $(x - x_0) \cdots (x - x_{k-1})$ is 1.

Now consider the Newton's divided-difference form starting with x_1:

$$Q(x) = f[x_1] + f[x_1, x_2](x - x_1) + \cdots + f[x_1, \ldots, x_k, x_0](x - x_1) \cdots (x - x_{k-1}),$$

where the points are relabeled as $y_0 = x_1$, $y_1 = x_2$, …, $y_{k-1} = x_k$, $y_k = x_0$. The coefficient of x^k in $Q(x)$ is $f[x_1, \ldots, x_k, x_0]$.

Since both $P(x)$ and $Q(x)$ are the unique interpolating polynomial of degree at most k for the same set of points, $P(x) = Q(x)$ for all x. Therefore, their leading coefficients are equal, so $f[x_1, \ldots, x_k, x_0] = f[x_0, x_1, \ldots, x_k]$.

Problem 4: Problems from the textbook, chapter 3.3

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(a)

Divided-Difference Table:

x_i	Zeroth	First	Second	Third
0	1			
		2.59488		
0.25	1.64872		3.36672	
		4.27824		2.91211
0.5	2.71828		5.55080	
		7.05364		
0.75	4.48169			

The coefficients for Newton's form are the leading divided differences: $f[x_0] = 1$, $f[x_0, x_1] = 2.59488$, $f[x_0, x_1, x_2] = 3.36672$, $f[x_0, x_1, x_2, x_3] = 2.91211$.

- **Degree 1 polynomial** (interpolates first 2 points):
  $P_1(x) = 1 + 2.59488(x - 0)$
  To find $P_1(0.43)$: Substitute $x = 0.43$ into the formula.
  $P_1(0.43) = 1 + 2.59488 \times 0.43 = 2.11580$
- **Degree 2 polynomial** (interpolates first 3 points):
  $P_2(x) = 1 + 2.59488(x - 0) + 3.36672(x - 0)(x - 0.25)$
  To find $P_2(0.43)$: Build nested from degree 1.
  $P_2(0.43) = P_1(0.43) + 3.36672 \times 0.43 \times (0.43 - 0.25) = 2.11580 + 3.36672 \times 0.43 \times 0.18 = 2.37638$
- **Degree 3 polynomial** (interpolates all 4 points):
  $P_3(x) = 1 + 2.59488(x - 0) + 3.36672(x - 0)(x - 0.25) + 2.91211(x - 0)(x - 0.25)(x - 0.5)$
  To find $P_3(0.43)$: Build nested from degree 2.
  $P_3(0.43) = P_2(0.43) + 2.91211 \times 0.43 \times 0.18 \times (0.43 - 0.5) = 2.37638 + 2.91211 \times 0.43 \times 0.18 \times (-0.07) = 2.36060$

(b)

Divided-Difference Table:

x_i	Zeroth	First	Second	Third
-0.5	1.93750			
		-2.42188		
-0.25	1.33203		1.81251	
		-1.06250		-1.00001
0.25	0.800781		0.81250	
		-0.45312		
0.5	0.687500			

The coefficients: $f[x_0] = 1.93750$, $f[x_0, x_1] = -2.42188$, $f[x_0, x_1, x_2] = 1.81251$, $f[x_0, x_1, x_2, x_3] = -1.00001$.

- **Degree 1 polynomial** (interpolates first 2 points):
  $P_1(x) = 1.93750 - 2.42188(x + 0.5)$
  To find $P_1(0)$: Substitute $x = 0$.
  $P_1(0) = 1.93750 - 2.42188 \times 0.5 = 0.72656$
- **Degree 2 polynomial** (interpolates first 3 points):
  $P_2(x) = 1.93750 - 2.42188(x + 0.5) + 1.81251(x + 0.5)(x + 0.25)$
  To find $P_2(0)$: Build nested from degree 1.
  $P_2(0) = P_1(0) + 1.81251 \times 0.5 \times 0.25 = 0.72656 + 1.81251 \times 0.125 = 0.95312$
- **Degree 3 polynomial** (interpolates all 4 points):
  $P_3(x) = 1.93750 - 2.42188(x + 0.5) + 1.81251(x + 0.5)(x + 0.25) - 1.00001(x + 0.5)(x + 0.25)(x - 0.25)$
  To find $P_3(0)$: Build nested from degree 2.
  $P_3(0) = P_2(0) - 1.00001 \times 0.5 \times 0.25 \times (-0.25) = 0.95312 - 1.00001 \times 0.125 \times (-0.25) = 0.98437$

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(a)

The data points are equally spaced with $h = 0.25$. Start by constructing the forward difference table:

x_i	$f(x_i)$	First differences (Δ)	Second differences (Δ^2)	Third differences (Δ^3)
0.00	1.00000			
		0.64872		
0.25	1.64872		0.42084	
		1.06956		0.27301
0.50	2.71828		0.69385	
		1.76341		
0.75	4.48169			

Here, $\Delta f(0) = 0.64872$, $\Delta^2 f(0) = 0.42084$, and $\Delta^3 f(0) = 0.27301$.

The Newton forward-difference formula is:

$$P_n(x) = f(0) + \sum_{k=1}^{n} \binom{s}{k} \Delta^k f(0),$$

where $s = \frac{x-0}{0.25} = 4x$.

- **Degree 1 polynomial**:

$$P_1(x) = 1 + \binom{4x}{1}(0.64872) = 1 + 4x \cdot 0.64872 = 1 + 2.59488x.$$

  To approximate $f(0.43)$, substitute $x = 0.43$ ($s = 1.72$):
  $P_1(0.43) = 2.11580$.
- **Degree 2 polynomial**:

$$P_2(x) = P_1(x) + \binom{4x}{2}(0.42084) = 1 + 2.59488x + \frac{4x(4x-1)}{2} \cdot 0.42084 = 1 + 2.59488x + 0.21042 \cdot (16x^2 - 4x).$$

  Simplifying the quadratic term: $0.21042 \cdot 16x^2 = 3.36672x^2$, $0.21042 \cdot (-4x) = -0.84168x$.
  So, $P_2(x) = 1 + (2.59488 - 0.84168)x + 3.36672x^2 = 1 + 1.7532x + 3.36672x^2$.
  $P_2(0.43) = 2.37638$.
- **Degree 3 polynomial**:

$$P_3(x) = P_2(x) + \binom{4x}{3}(0.27301) = P_2(x) + \frac{4x(4x-1)(4x-2)}{6} \cdot 0.27301.$$

  The cubic term simplifies to $\frac{0.27301}{6} \cdot 4x(4x-1)(4x-2) = 0.04550167 \cdot (64x^3 - 48x^2 + 8x)$.
  So, $0.04550167 \cdot 64x^3 = 2.912107x^3$, $0.04550167 \cdot (-48x^2) = -2.18408x^2$, $0.04550167 \cdot 8x = 0.364013x$.
  Combining with $P_2(x)$: $P_3(x) = 1 + (1.7532 + 0.364013)x + (3.36672 - 2.18408)x^2 + 2.912107x^3 = 1 + 2.117213x + 1.18264x^2 + 2.912107x^3$.
  $P_3(0.43) = 2.36060$.

(b)

The data points are equally spaced with $h = 0.1$. The forward difference table is:

x_i	$f(x_i)$	First differences (Δ)	Second differences (Δ^2)	Third differences (Δ^3)
0.10	-0.29004986			
		-0.27074748		
0.20	-0.56079734		0.01752510	
		-0.25322238		-0.00291320
0.30	-0.81401972		0.01461190	
		-0.23861048		
0.40	-1.05263020			

Here, $\Delta f(0.1) = -0.27074748$, $\Delta^2 f(0.1) = 0.01752510$, and $\Delta^3 f(0.1) = -0.00291320$.

The Newton forward-difference formula is:

$$P_n(x) = f(0.1) + \sum_{k=1}^{n} \binom{s}{k} \Delta^k f(0.1),$$

where $s = \frac{x-0.1}{0.1} = 10(x-0.1)$.

- **Degree 1 polynomial**:

$$P_1(x) = -0.29004986 + \binom{10(x-0.1)}{1}(-0.27074748) = -0.29004986 + 10(x-0.1) \cdot (-0.27074748) = -0.29004986 - 2.7074748(x-0.1).$$

  $P_1(0.18) = -0.50664784$.
- **Degree 2 polynomial**:

$$P_2(x) = P_1(x) + \binom{10(x-0.1)}{2}(0.01752510) = P_1(x) + \frac{10(x-0.1)\cdot(10(x-0.1)-1)}{2}\cdot 0.01752510.$$

The quadratic term: $\frac{0.01752510}{2}\cdot 10(x-0.1)(10(x-0.1)-1) = 0.00876255\cdot(100(x-0.1)^2 - 10(x-0.1))$.
So, $0.00876255\cdot 100(x-0.1)^2 = 0.876255(x-0.1)^2$, $0.00876255\cdot(-10(x-0.1)) = -0.0876255(x-0.1)$.
Combining:
$P_2(x) = -0.29004986 + (-2.7074748 - 0.0876255)(x-0.1) + 0.876255(x-0.1)^2 = -0.29004986 - 2.7951003(x-0.1) + 0.876255(x-0.1)^2$.
$P_2(0.18) = -0.50804985$.

- **Degree 3 polynomial**:

$$P_3(x) = P_2(x) + \binom{10(x-0.1)}{3}(-0.00291320) = P_2(x) + \frac{10(x-0.1)\cdot(10(x-0.1)-1)\cdot(10(x-0.1)-2)}{6}\cdot(-0.00291320).$$

The cubic term: $\frac{-0.00291320}{6}\cdot 10(x-0.1)(10(x-0.1)-1)(10(x-0.1)-2) = -0.000485533\cdot(1000(x-0.1)^3 - 300(x-0.1)^2 + 20(x-0.1))$.
So, $-0.000485533\cdot 1000(x-0.1)^3 = -0.485533(x-0.1)^3$, $-0.000485533\cdot(-300(x-0.1)^2) = 0.14566(x-0.1)^2$,
$-0.000485533\cdot 20(x-0.1) = -0.00971066(x-0.1)$.
Combining with $P_2(x)$:
$P_3(x) = -0.29004986 + (-2.7951003 - 0.00971066)(x-0.1) + (0.876255 + 0.14566)(x-0.1)^2 - 0.485533(x-0.1)^3 = -0.29004986 - 2.80481096(x-0.1) +$
.
$P_3(0.18) = -0.50814307$.

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(a)

Nodes in order:
$x_0 = -0.1, x_1 = 0, x_2 = 0.2, x_3 = 0.3$ with
$f(x_0) = 5.30000, f(x_1) = 2.00000, f(x_2) = 3.19000, f(x_3) = 1.00000$.

Divided-difference table (key entries):

- Zeroth:
  $f[x_0] = 5.3, f[x_1] = 2.0, f[x_2] = 3.19, f[x_3] = 1.0$
- First:
  $f[x_0, x_1] = -33.0, f[x_1, x_2] = 5.95, f[x_2, x_3] = -21.9$
- Second:
  $f[x_0, x_1, x_2] = \dfrac{779}{6} \approx 129.833333, f[x_1, x_2, x_3] = -\dfrac{557}{6} \approx -92.833333$
- Third:
  $f[x_0, x_1, x_2, x_3] = -\dfrac{1670}{3} \approx -556.666667$

Newton form:

$$P_3(x) = 5.3 - 33(x+0.1) + \frac{779}{6}(x+0.1)(x) - \frac{1670}{3}(x+0.1)(x)(x-0.2)$$

Expanded:

$$P_3(x) = -\frac{1670}{3}x^3 + \frac{371}{2}x^2 - \frac{533}{60}x + 2$$

(b)

Add $x_4 = 0.35, f(x_4) = 0.97260 = 4863/5000$.

New (only the needed) divided differences:

- The new top-of-column coefficient is

$$f[x_0, x_1, x_2, x_3, x_4] = \frac{516016}{189} \approx 2730.243386.$$

Thus the degree-4 Newton form is

$$P_4(x) = P_3(x) + \frac{516016}{189}(x+0.1)(x)(x-0.2)(x-0.3)$$

Expanded:

$$P_4(x) = \frac{516016}{189}x^4 - \frac{1558082}{945}x^3 + \frac{2010983}{9450}x^2 + \frac{236191}{31500}x + 2$$

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Because P_3 is the Newton interpolating polynomial through the nodes $0, 0.25, 0.5, 0.75$, it satisfies $P_3(x_k) = f(x_k)$. So $f(0.75) = P_3(0.75)$.

Compute:

$$P_3(0.75) = 1 + 4(0.75) + 4(0.75)(0.75 - 0.25) + \frac{16}{3}(0.75)(0.75 - 0.25)(0.75 - 0.5) = 6.$$

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The degree can be tested using finite differences, as the nodes are equally spaced (−2,−1,0,1,2,3 with step $h = 1$). For a polynomial P of degree n, the n -th forward differences are constant (and the $(n + 1)$ -st are zero).

The difference table for the given data is:

x	-2	-1	0	1	2	3
f	1	4	11	16	13	-4
Δf		3	7	5	-3	-17
$\Delta^2 f$			4	-2	-8	-14
$\Delta^3 f$				-6	-6	-6
$\Delta^4 f$					0	0

- The third differences are constant and nonzero: $\Delta^3 f \equiv -6$.
- The fourth differences are zero.

Therefore, the unique interpolating polynomial that fits these six points has degree exactly 3 (cubic).

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The given equation follows directly from combining the interpolation error formula in Eq. (3.3) with the Newton divided-difference form of the interpolating polynomial.

From Eq. (3.3), for the interpolating polynomial $P_n(x)$ of degree at most n that agrees with f at the distinct points $x_0, \ldots, x_n$,

$$f(x) = P_n(x) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!} \prod_{i=0}^{n} (x - x_i),$$

where $\xi(x)$ lies between $\min\{x_0, \ldots, x_n, x\}$ and $\max\{x_0, \ldots, x_n, x\}$.

Now consider the Newton divided-difference formula for the interpolating polynomial of degree at most $n + 1$ that agrees with f at the points $x_0, \ldots, x_n, x$. This polynomial, denoted $P_{n+1}(y)$, satisfies $P_{n+1}(x) = f(x)$ and can be expressed as

$$P_{n+1}(y) = P_n(y) + f[x_0, \ldots, x_n, x] \prod_{i=0}^{n} (y - x_i).$$

Evaluating at $y = x$,

$$f(x) = P_{n+1}(x) = P_n(x) + f[x_0, \ldots, x_n, x] \prod_{i=0}^{n} (x - x_i).$$

Equating the two expressions for $f(x)$,

$$P_n(x) + f[x_0, \ldots, x_n, x] \prod_{i=0}^{n} (x - x_i) = P_n(x) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!} \prod_{i=0}^{n} (x - x_i).$$

Subtracting $P_n(x)$ from both sides and dividing by $\prod_{i=0}^{n} (x - x_i)$ (assuming $x \neq x_i$ for all i),

$$f[x_0, \ldots, x_n, x] = \frac{f^{(n+1)}(\xi(x))}{(n+1)!}.$$

Problem 5: Hermite Interpolation

```
function coeffs = hermite(xi, fi, dfi)
    % basic checks
    if ~(isvector(xi) && isvector(fi) && isvector(dfi))
        error('xi, fi, dfi must be vectors.');
    end
    xi  = xi(:);  fi  = fi(:);  dfi = dfi(:);
```

```
    n = numel(xi);
    if numel(fi) ~= n || numel(dfi) ~= n
        error('xi, fi, dfi must have the same length.');
    end
    if numel(unique(xi)) ~= n
        error('Entries of xi must be distinct.');
    end

    % build duplicated node list z = [x0 x0 x1 x1 ... xn xn]
    m = 2*n;               % total rows in Hermite table
    z = zeros(m,1);
    Q = zeros(m,m);        % divided-difference table

    for i = 1:n
        z(2*i-1) = xi(i);
        z(2*i  ) = xi(i);

        % first column (function values)
        Q(2*i-1,1) = fi(i);
        Q(2*i  ,1) = fi(i);

        % first divided differences:
        % for the duplicated pair, replace the 0/0 with f'(xi)
        Q(2*i,2) = dfi(i);  % f[z_{2i-1}, z_{2i}] = f'(xi)

        % the other first-difference in between distinct nodes is standard
        if i > 1
            Q(2*i-1,2) = (Q(2*i-1,1) - Q(2*i-2,1)) / (z(2*i-1) - z(2*i-2));
        end
    end

    % higher-order divided differences (same Newton recursion)
    for j = 3:m
        for i = j:m
            Q(i,j) = (Q(i,j-1) - Q(i-1,j-1)) / (z(i) - z(i-j+1));
        end
    end

    % Newton coefficients are the diagonal of the table
    coeffs = diag(Q).';
end
```